DFT

- Foward Fourier Transform

$$X(jw) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt}\, dt$$

- Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(jw)\, e^{jwt}\, dw$$

} continuous time continuous f

- Fourier series coefficient

$$a_k = \frac{1}{T_0} \int_{0}^{T_0} x(t)\, e^{-j 2\pi f_0 k t}\, dt$$

- Fourier series

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j 2\pi f_0 k t}$$

} continuous time discrete f

- FIR filter Frequency response

$$H(e^{j\hat{w}}) = \sum_{k=0}^{M} h[k] e^{-j\hat{w}k}$$

} discrete time continuous f

- Discrete time Fourier transform

$$X(e^{j\hat{w}}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\hat{w}n}$$

} discrete time continuous f

- Discrete Fourier Transform

$$X_N(e^{j\hat{\omega}_k}) = \sum_{n=0}^{N-1} x_N[n] e^{\frac{-j2\pi k}{N} n}$$

$0 \le k \le N-1$

discrete time
discrete f

$$\hat{\omega}_k = \frac{2\pi k}{N}$$

$$\underbrace{X_N(e^{j\frac{2\pi k}{N}})}_{\| \; X_N[k]} = \sum_{n=0}^{N-1} x_N[n] e^{\frac{-j2\pi k}{N} n}$$

forward DFT

$$X_N[k] = \sum_{n=0}^{N-1} x_N[n] e^{-j\frac{2\pi k}{N} n} \qquad 0 \le k \le N-1$$

inverse DFT

$$x_N[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_N[k] e^{j\frac{2\pi k}{N} n} \qquad 0 \le n \le N-1$$

- 예제

$N=4$, $x[0]=1$, $x[1]=2$, $x[2]=3$, $x[3]=4$

$$X[k] = \sum_{n=0}^{3} x[n] e^{-\frac{2\pi k}{N} n}$$

$k=0$ 일 때 $X[0] = 1 \times e^{\frac{-2\pi \times 0}{4} \times 0} + 2 \times e^{\frac{-2\pi \times 0}{4} \times 1}$

$$3\times e^{\frac{-2\pi\times 0}{4}\times 2} + 4\times e^{\frac{-2\pi\times 0}{4}\times 3}$$

$$= 1\times e^{0} + 2\times e^{0} + 3\times e^{0} + 4\times e^{0} = 1+2+3+4 = 10$$

k=1일때 $X[1] = 1\times e^{\frac{-2\pi\times 1}{4}\times 0} + 2\times e^{\frac{-2\pi\times 1}{4}\times 1}$

$$+ 3\times e^{\frac{-2\pi\times 1}{4}\times 2} + 4\times e^{\frac{-2\pi\times 1}{4}\times 3}$$

$$= 1\times e^{0} + 2\times e^{-\frac{\pi}{2}} + 3\times e^{-\pi} + 4\times e^{-\frac{6}{4}\pi}$$

$= e^{\frac{1}{2}\pi}$

$$= 1 + 2(-j) + 3(-1) + 4(j)$$

$$= -2 + j2$$

k=2일때

$$X[2] = \sum_{n=0}^{3} x[n] e^{-\frac{2\pi\times 2}{4}n} = \sum_{n=0}^{3} x[n] e^{-\pi n}$$

$$= x[0]e^{-0} + x[1]e^{-\pi} + x[2]e^{-2\pi} + x[3]e^{-3\pi}$$

$$= 1\times 1 + 2(-1) + 3(1) + 4(-1)$$

$$= 1 - 2 + 3 - 4 = -2$$

k=3일때 $X[3] = \sum_{n=0}^{3} x[n] e^{-\frac{2\pi\times 3}{4}n} = \sum_{n=0}^{3} x[n] e^{-\frac{3}{2}\pi n}$

$$x[0]e^{-0} + x[1]e^{-\frac{3}{2}\pi} + x[2]e^{-\frac{6}{2}\pi} + x[3]e^{-\frac{9}{2}\pi}$$

$1 \times 1 + 2 \times (j) + 3 \times (-1) + 4 \times (-j)$

$= 1 + j2 - 3 - j4 = -2 - j2$